SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
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॥वसुधैव कुटुम्बकम्॥

CSE - Academic Year 2026-27
Java Programming Lab | Batch 2025-29

ASSIGNMENT : 24

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Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Develop a Java GUI application integrated with JDBC to perform database operations.

Learning Objectives : To integrate Java GUI applications with databases using JDBC.

Theory : Java Swing provides GUI components such as buttons, labels, text fields, tables, and forms. JDBC connects a Java application to a database. Combining Swing and JDBC allows users to enter, store, retrieve, and manage records through an interactive desktop interface.

Exercise :

1. Create a Library Management System GUI connected with a database.

```
import java.awt.*;
```

```
import java.sql.*;
import javax.swing.*;

public class LibraryGUI {
    public static void main(String[] args) {
        JFrame f = new JFrame("Library Management System");

        JLabel l1 = new JLabel("Book ID:");
        JLabel l2 = new JLabel("Book Name:");

        JTextField t1 = new JTextField(15);
        JTextField t2 = new JTextField(15);

        JButton b1 = new JButton("Add Book");
        JButton b2 = new JButton("View Books");

        JTextArea area = new JTextArea(10, 30);
        area.setEditable(false);

        f.setLayout(new FlowLayout(FlowLayout.CENTER, 15, 10));

        f.add(l1);
        f.add(t1);
        f.add(l2);
        f.add(t2);
        f.add(b1);
        f.add(b2);
        f.add(new JScrollPane(area));

        b1.addActionListener(e -> {
            try {
                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/college",
                    "root", "anushree");

                String sql = "insert into books values(?, ?)";
                PreparedStatement ps = con.prepareStatement(sql);

                ps.setInt(1, Integer.parseInt(t1.getText()));
                ps.setString(2, t2.getText());
```

```
                ps.executeUpdate();
                area.setText("Book added successfully!");

                ps.close();
                con.close();
            } catch (Exception ex) {
                area.setText("Error: " + ex.getMessage());
            }
        });

        b2.addActionListener(e -> {
            try {
                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/college",
                    "root", "anushree");

                Statement st = con.createStatement();
                ResultSet rs = st.executeQuery("select * from books");

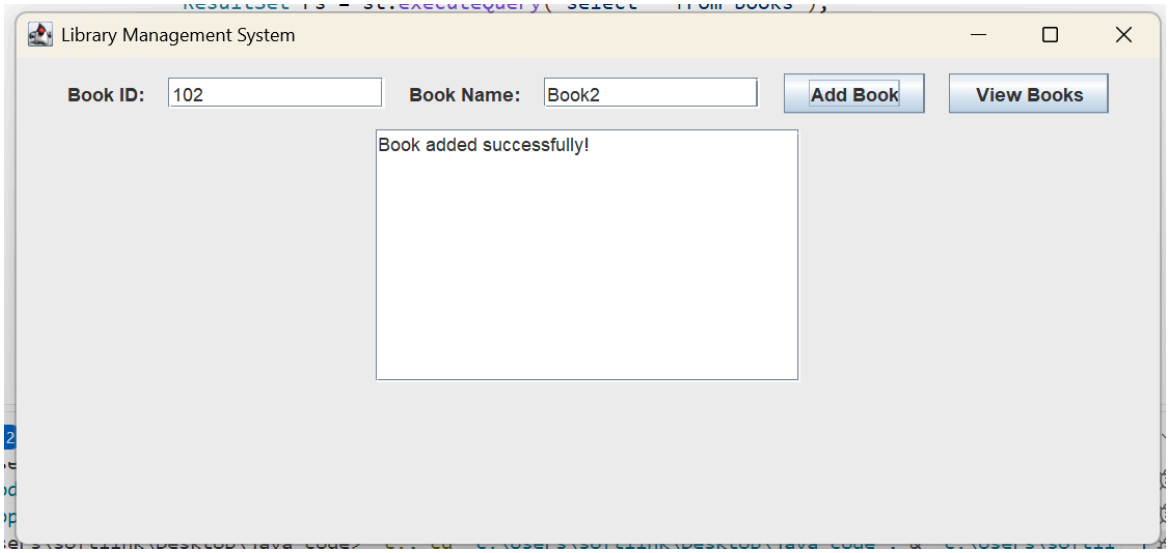
                area.setText("Book ID\tBook Name\n");
                area.append("-----------------------------\n");

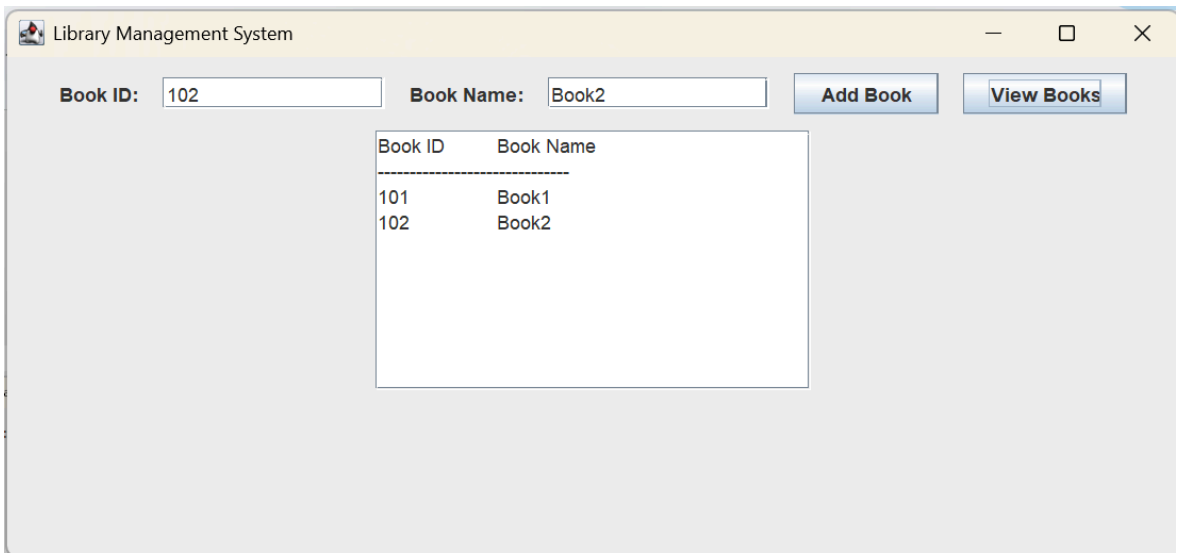
                while (rs.next()) {
                    area.append(rs.getInt(1) + "\t");
                    area.append(rs.getString(2) + "\n");
                }
                rs.close();
                st.close();
                con.close();
            } catch (Exception ex) {
                area.setText("Error: " + ex.getMessage());
            }
        });
        f.setSize(420, 350);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f.setLocationRelativeTo(null);
        f.setVisible(true);
    }
}
```

In SQL :

```
10 • use college;
11 • create table books(book_id int primary key, title varchar(100));
```

OUTPUT :





2. Create a Book Issue Tracking System GUI connected with a database.

```
import java.awt.*;
import java.sql.*;
import javax.swing.*;
```

```
public class BookIssueGUI {
    public static void main(String[] args) {
        JFrame f = new JFrame("Book Issue Tracking System");
        JTextField t1 = new JTextField(15);
        JTextField t2 = new JTextField(15);
        JTextField t3 = new JTextField(15);
        JTextField t4 = new JTextField(15);

        JButton b1 = new JButton("Issue Book");
        JButton b2 = new JButton("View Records");
        JTextArea area = new JTextArea(10, 30);
        area.setEditable(false);

        f.setLayout(new FlowLayout(FlowLayout.CENTER, 15, 8));
        f.add(new JLabel("Book ID:"));
        f.add(t1);
        f.add(new JLabel("Student Name:"));
        f.add(t2);
        f.add(new JLabel("Issue Date (YYYY-MM-DD):"));
        f.add(t3);
        f.add(new JLabel("Return Date (YYYY-MM-DD):"));
        f.add(t4);
        f.add(b1);
        f.add(b2);
        f.add(new JScrollPane(area));

        b1.addActionListener(e -> {
            try {
                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/college",
                    "root", "anushree");

                String sql = "insert into book_issue values(?, ?, ?,
?)";
                PreparedStatement ps = con.prepareStatement(sql);
                ps.setInt(1, Integer.parseInt(t1.getText()));
                ps.setString(2, t2.getText());
                ps.setDate(3, Date.valueOf(t3.getText()));
                ps.setDate(4, Date.valueOf(t4.getText()));
                ps.executeUpdate();
```

```
                area.setText("Book issue record saved successfully!");
                ps.close();
                con.close();
            } catch (Exception ex) {
                area.setText("Error: " + ex.getMessage());
            }
        });
        b2.addActionListener(e -> {
            try {
                Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/college",
                    "root", "password");

                Statement st = con.createStatement();
                ResultSet rs = st.executeQuery("select * from
book_issue");

                area.setText("Book ID | Student | Issue Date | Return
Date\n");

area.append("-------------------------------------------------\n");

                while (rs.next()) {
                    area.append(
                        rs.getInt(1) + " | " +
                        rs.getString(2) + " | " +
                        rs.getDate(3) + " | " +
                        rs.getDate(4) + "\n"
                    );
                }

                rs.close();
                st.close();
                con.close();
            } catch (Exception ex) {
                area.setText("Error: " + ex.getMessage());
            }
        });

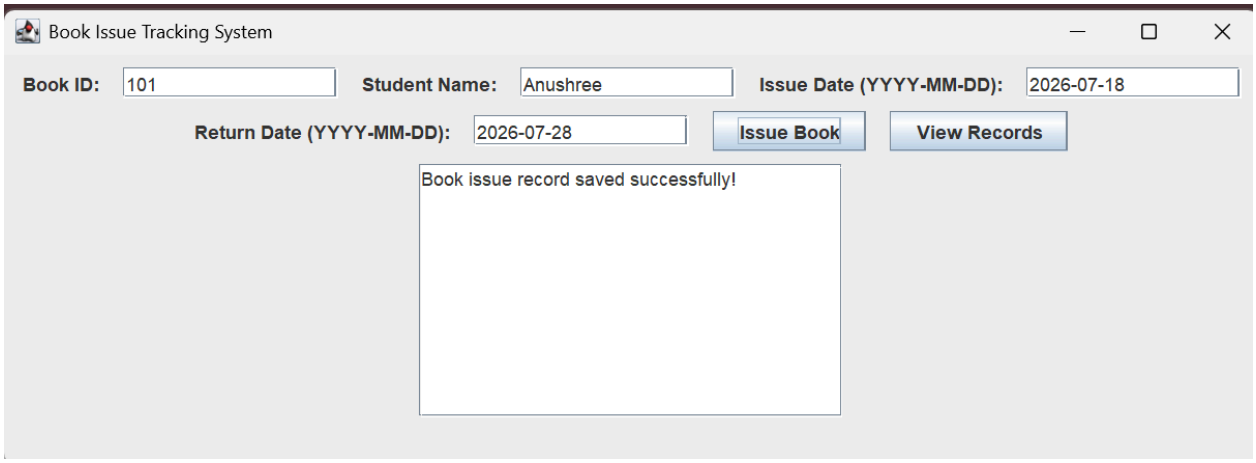
        f.setSize(480, 500);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
```

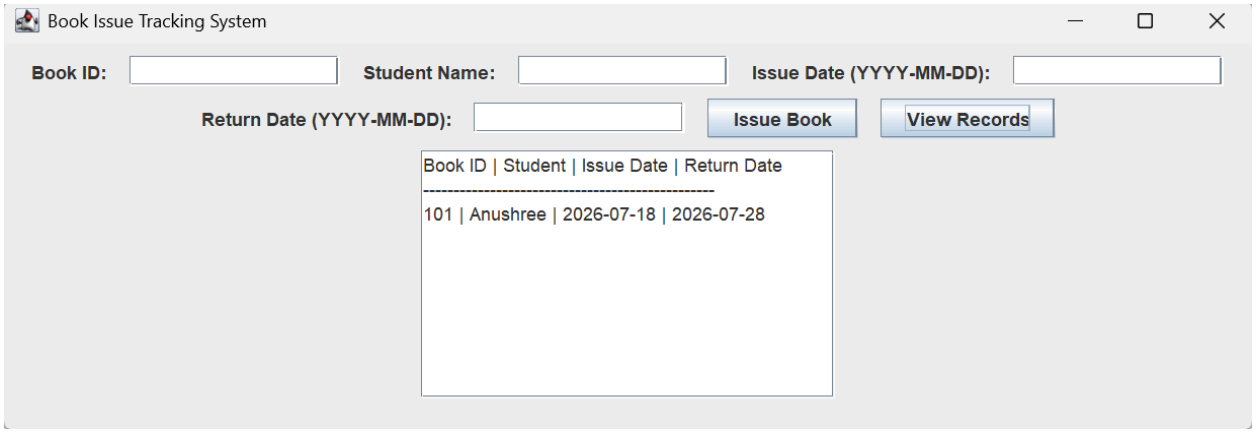
```
        f.setLocationRelativeTo(null);
        f.setVisible(true);
    }
}
```

In SQL :

```
13 •    create table book_issue ( book_id int, student_name varchar(60), issue_date date, return_date date );
```

OUTPUT :





Outcome :

- Understood the integration of Java Swing and JDBC for database operations.
- Learned to create graphical user interfaces using JFrame, JTextField, JButton, and JTextArea.
- Understood how to connect a Java application to a MySQL database.
- Learned to insert and store records using PreparedStatement and executeUpdate().
- Successfully implemented GUI applications for library management and book issue tracking.

Conclusion :

This assignment demonstrated the integration of Java Swing and JDBC to develop interactive database applications. Library book details and book issue records were managed through graphical interfaces connected to a MySQL database. The assignment helped in understanding GUI component handling, database connectivity, and executing SQL operations through a Java desktop application.